

TatvaOps AI-Powered Omnichannel Lead Qualification System

Project Documentation

1. Introduction

TatvaOps AI Omnichannel System is an AI-powered customer engagement and lead qualification platform designed specifically for the construction and real-estate industry. The platform uses WhatsApp as the primary communication channel to automate customer enquiries, intelligently route users to specialized AI consultants, collect structured lead information, and provide summarized insights to the sales team through an admin dashboard.

The platform acts as a bridge between:

- Customers
- Sales teams
- Vendors
- Construction service providers

The system aims to simplify customer communication, improve lead management, and increase conversion efficiency using AI-driven automation.

2. Problem Statement

Construction and real-estate businesses often face challenges such as:

- Missed customer enquiries
- Delayed responses
- Manual lead handling
- Poor follow-up systems
- Unorganized communication channels
- Difficulty tracking customer requirements
- Inefficient sales workflows

Leads generated through:

- Instagram reels
- Sponsored ads
- Social media campaigns
- WhatsApp enquiries

are often not properly qualified or tracked.

TatvaOps solves this problem by creating an AI-powered omnichannel lead qualification and customer management system.

3. Project Objectives

The objectives of the proposed system are:

- Automate customer enquiry handling through WhatsApp
 - Provide instant AI-powered customer interaction
 - Route users to specialized AI consultants based on services
 - Collect structured lead information using hybrid conversational flows
 - Support MCQ, descriptive responses, and file attachments
 - Generate AI-based customer requirement summaries
 - Store all enquiry details in a centralized admin dashboard
 - Help the sales team identify and prioritize leads
 - Improve customer engagement and response time
 - Build a scalable omnichannel architecture for future expansion
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4. Proposed System Overview

The system combines:

- WhatsApp communication
- Twilio APIs
- AI-powered conversational agents
- Intelligent service routing
- Structured lead qualification workflows
- Admin CRM dashboard

to create a smart lead qualification and customer engagement platform.

5. Service Categories & AI Consultants

The platform contains multiple AI consultants specialized in different construction and home service domains.

Service	AI Consultant
Residential Construction	Aravind Narayanan
Home Interiors	Aadhya
Painting & Waterproofing	Manjunath Gowda
Electrical Services	Vivek Shetty
Solar Rooftop Installation	Kavya Nair
Home Automation	Riya Mehta

Each consultant acts as a specialized conversational AI agent responsible for handling enquiries related to their domain expertise.

6. System Workflow

Step 1 — Lead Generation

Customers interact with:

- Instagram reels
- Sponsored advertisements
- Social media campaigns

The advertisement contains:

Enquire on WhatsApp

When the user clicks the button, WhatsApp opens automatically.

Step 2 — Main AI Agent Interaction

The main AI agent named **Nova** welcomes the user and introduces TatvaOps.

Example:

Hi 🙌

Welcome to TatvaOps.

I'm Nova, your smart construction assistant.

We help with:

1. Residential Construction
2. Home Interiors
3. Painting & Waterproofing
4. Electrical Services
5. Solar Rooftop Installation
6. Home Automation

Please reply with the service you're looking for.

Nova acts as:

- AI receptionist
 - Intent classifier
 - Service router
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Step 3 — Intelligent Service Routing

Based on the user's selected service, Nova routes the user to a specialized AI consultant.

Example:

- Residential Construction → Aravind Narayanan
 - Home Interiors → Aadhya
 - Painting & Waterproofing → Manjunath Gowda
 - Electrical Services → Vivek Shetty
 - Solar Rooftop Installation → Kavya Nair
 - Home Automation → Riya Mehta
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7. Hybrid Conversational Flow

The system uses a hybrid conversational model consisting of:

- MCQ-based questions
- Descriptive questions
- File attachment uploads

This allows the platform to collect both structured and detailed customer requirements.

7.1 MCQ-Based Questions

The AI consultant asks structured multiple-choice questions for faster and more organized lead qualification.

Example:

What type of property do you have?

1. Apartment
2. Independent House
3. Villa
4. Commercial Building

Example:

What is your estimated budget range?

1. Below 5 Lakhs
 2. 5–10 Lakhs
 3. 10–20 Lakhs
 4. Above 20 Lakhs
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7.2 Descriptive Questions

The AI consultant may ask users to describe their requirements in detail.

Example:

Please describe your preferred interior style or specific requirements.

Descriptive responses help collect:

- Customer preferences
 - Design expectations
 - Custom requirements
 - Additional project information
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7.3 File Attachment Questions

The system supports customer file uploads through WhatsApp.

Customers can upload:

- Floor plans
- Site images
- Reference designs
- Requirement documents
- Property photos

Example:

Please upload your floor plan, site image, or reference design if available.

These files are stored and linked to the customer enquiry in the admin dashboard.

8. Lead Qualification Flow

The AI consultant collects:

- Property type
- Budget range
- Timeline
- Location
- Service preferences
- Project description
- Reference images/documents

After completion:

- AI generates a summarized report
 - Structured lead fields are extracted
 - Customer files are linked
 - Lead data is stored in the CRM dashboard
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9. Summary Generation

Example summary:

Customer requires home interior services for a 3BHK apartment in Bengaluru with a budget range of 10–15 Lakhs. The customer prefers a modern minimalist interior style and plans to begin within 2 months. Reference images and floor plan have been uploaded.

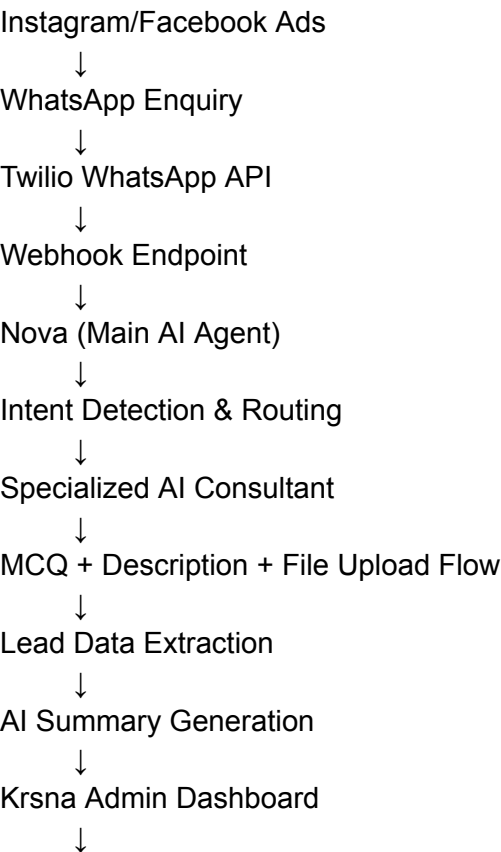
10. Krsna Admin Dashboard

All enquiry details are stored in the Krsna Admin Panel.

The dashboard allows the operations and sales team to:

- Monitor live conversations
 - View enquiry summaries
 - Read structured customer responses
 - Access uploaded files
 - Analyze lead quality
 - Track sessions
 - Contact customers for sales follow-up
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11. System Architecture



12. Main Modules

12.1 WhatsApp Communication Module

Handles:

- WhatsApp messaging
- Incoming webhook requests
- Session creation
- Media handling
- File uploads

Technology Used:

- Twilio WhatsApp API
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12.2 Main AI Agent (Nova)

Responsibilities:

- Greeting users
 - Explaining available services
 - Understanding user intent
 - Routing users to specialized AI consultants
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12.3 Specialized AI Consultants

Aravind Narayanan — Residential Construction Consultant

Handles:

- House construction enquiries
- Construction planning

- Structural consultation
 - Timeline estimation
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Aadhya — Home Interior Consultant

Handles:

- Interior design requirements
 - Style preferences
 - Space planning
 - Furniture and layout consultation
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Manjunath Gowda — Painting & Waterproofing Consultant

Handles:

- Painting services
 - Waterproofing solutions
 - Exterior and interior coatings
 - Surface treatment consultation
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Vivek Shetty — Electrical Services Consultant

Handles:

- Electrical installation
 - Wiring solutions
 - Smart electrical systems
 - Maintenance services
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Kavya Nair — Solar Rooftop Installation Consultant

Handles:

- Solar panel installation
- Rooftop solar consultation
- Energy-saving systems

- Solar project requirements
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Riya Mehta — Home Automation Consultant

Handles:

- Smart home solutions
 - IoT-based home automation
 - Security automation
 - Smart lighting and appliance integration
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12.4 Conversation Engine

This module manages:

- Structured conversation stages
 - MCQ handling
 - Descriptive responses
 - File attachment workflows
 - Lead qualification logic
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12.5 Summary & Lead Intelligence Module

Responsibilities:

- AI-generated summaries
 - Structured field extraction
 - Lead categorization
 - Lead scoring
 - Requirement analysis
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12.6 Krsna Admin Dashboard

Features:

- Dashboard analytics
- Sessions page
- Enquiries page
- Summaries page
- Logs page
- Uploaded file management
- System health monitoring

The dashboard acts as the operational control center for the platform.

13. Technology Stack

Component	Technology
Frontend	React.js + Vite
Backend	Python + FastAPI
Communication API	Twilio WhatsApp API
AI Integration	Google Gemini API
Database	Supabase PostgreSQL
Session Store	Redis / Upstash
Styling	Tailwind CSS
Deployment	Render / Railway
Development Tunnel	ngrok

14. Features of the System

- AI-powered WhatsApp communication
- Intelligent service routing
- Multi-agent conversational architecture
- MCQ-based lead qualification
- Descriptive customer interaction
- File attachment support
- AI-generated summaries

- Structured lead extraction
 - Real-time admin monitoring
 - CRM-based enquiry management
 - Sales workflow integration
 - Omnichannel-ready architecture
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15. Advantages of the Proposed System

- Faster customer engagement
 - Reduced manual workload
 - Better lead organization
 - Improved response time
 - Structured sales pipeline
 - Enhanced customer experience
 - Better lead qualification
 - Improved conversion rates
 - Centralized customer communication
 - Scalable architecture
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16. Future Enhancements

Future versions of the platform may include:

- Website landing page integration
 - Voice AI agents
 - Instagram DM integration
 - AI-generated quotations
 - Vendor marketplace system
 - Project milestone tracking
 - AI-based lead scoring
 - Human agent handoff system
 - Real-time analytics dashboard
 - Multi-language support
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17. Expected Outcome

The proposed system will:

- Automate customer enquiry management
 - Improve operational efficiency
 - Increase lead conversion rates
 - Centralize customer communication
 - Improve coordination between teams and vendors
 - Create a scalable AI-driven CRM ecosystem
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18. Conclusion

TatvaOps AI Omnichannel Lead Qualification System is an intelligent AI-powered customer engagement platform designed for the construction and real-estate industry. By combining WhatsApp communication, conversational AI agents, hybrid conversational workflows, AI-generated summaries, and centralized CRM management, the platform creates a scalable and efficient solution for modern construction businesses.

The system provides a strong foundation for future omnichannel expansion while delivering immediate business value through intelligent automation and structured lead management.